

SPL-01 Pilot Aseptic Syringe Filling Line

AAU SmartProduction Laboratory

Line Description and Equipment Register

This document describes the physical arrangement, installed equipment, and interconnections of the SPL-01 pilot aseptic syringe filling line located at the AAU SmartProduction Laboratory. It is intended for facilities management, maintenance personnel, and system integrators who need a single reference for the line's configuration.

It does not replace the individual equipment lifecycle or interface documents; those are listed in the document register at the end of this file.

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About This Document

This document has been prepared by Elara Automation GmbH and is intended for qualified technical personnel responsible for the installation, commissioning, operation, and maintenance of the equipment described herein. Readers are expected to be familiar with general principles of industrial automation, electrical safety, and network-connected laboratory equipment.

The document is structured as a reference manual. Sections are broadly ordered to follow the typical lifecycle of the equipment — from first delivery through to end-of-life disposal — but individual sections may be read in isolation. Cross-references are provided where information in one section depends on material covered elsewhere.

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Conventions Used in This Document

The following conventions are used throughout:

Notation	Meaning
{serialNumber}	The unit's serial number as printed on the rear label (e.g. EA-PS-2024-00391)
{base}	The site-specific topic or namespace prefix, configured during commissioning
NOTE	Supplementary information that aids understanding but is not safety-critical
CAUTION	Risk of equipment damage or process disruption if instruction is not followed
WARNING	Risk of personal injury if instruction is not followed

1 Line Description

The SPL-01 line is a two-station pilot filling line designed for small-batch aseptic syringe filling and stoppering in a laboratory environment. The line operates at up to approximately 120 syringes per hour, limited in practice by the fill-cycle time of the upstream filling station and the manual syringe transfer between stations.

Both stations are manufactured by Elara Automation GmbH (Brückenstraße 14, 70173 Stuttgart, Germany) and belong to the company's LinFill and PlungerSet product families. They share a common electronics platform — ESP32 microcontroller, L298N motor driver, 12 V DC lift motor, and lead-screw assembly — but differ in their application-layer communication protocols: the filling station uses MQTT, the stoppering station uses OPC UA.

The line is hosted on an ACOPOS-compatible laboratory table, approximately 900 mm × 600 mm, with a shared 12 V DC power rail rated at 5 A. Both stations connect to the facility network via a 2.4 GHz WiFi access point mounted on the ceiling of the pilot filling area (PFA-01). The MQTT broker (Eclipse Mosquitto 2.x) and OPC UA client workstation are rack-mounted in the adjacent IT room.

2 Physical Layout

The two stations are positioned side-by-side on the table, left to right in process flow order. Station 1 (LinFill-120) occupies the left position; Station 2 (PlungerSet-80) is to its right, approximately 250 mm centre-to-centre. A manual carrier tray is used to transport syringes from Station 1 to Station 2 between process steps.

The filling needle of Station 1 is positioned directly above the syringe position in the carrier tray. Station 2 is oriented so that the stoppering head aligns with the same syringe position after the tray is slid laterally. Both stations are bolted to the table surface using M6 hardware on the 150 × 120 mm bolt-circle pattern common to both products.

Position	Station	Process Step	Protocol	Direction of Flow
Station 1 — left	LinFill-120 Syringe Filling Station	Vertical needle fill of pre-positioned syringe	MQTT 3.1.1	↓ Syringe tray transfers right
Station 2 — right	PlungerSet-80 Automated Stoppering Station	Rubber plunger feed and insertion into filled syringe	OPC UA TCP	→ Tray exits line after stoppering

3 Installed Equipment

The following table lists all fixed assets associated with the SPL-01 line. Item numbers are used as internal reference identifiers only and do not correspond to any external asset management scheme.

Item	Description	Manufacturer / Supplier	Order Code	Serial / ID
SPL-01-001	LinFill-120 Vertical Syringe Filling Station	Elara Automation GmbH	EA-LF120-230V-EU	EA-LF-2024-00472
SPL-01-002	PlungerSet-80 Automated Stoppering Station	Elara Automation GmbH	EA-PS080-230V-EU	EA-PS-2024-00391
SPL-01-003	Laboratory work table, 900 × 600 mm, ACOPOS-compatible surface	Facility-supplied	—	TABLE-PFA 01-003
SPL-01-004	12 V DC shared power rail, 5 A, IEC 60320 C14 inlet	Facility-supplied	—	PWR-PFA0 1-004

Item	Description	Manufacturer / Supplier	Order Code	Serial / ID
SPL-01-005	WiFi Access Point, 2.4 GHz 802.11 b/g/n	Facility IT	—	AP-PFA01-005
SPL-01-006	MQTT Broker — Eclipse Mosquitto 2.x	Open source (Eclipse Foundation)	—	Server: SRV-IT01 (rack)
SPL-01-007	OPC UA Client / SCADA workstation	Facility IT	—	WS-CTRL-007

4 Component Hierarchy

The component hierarchy below lists each station's sub-assemblies as documented in the individual product datasheets. Both stations share the lift sub-assembly design; spare parts for the 12 V DC lift motor and lead-screw assembly are interchangeable.

Station 1 — LinFill-120 (EA-LF-2024-00472)

Sub-assembly	Key Specification
ESP32 Controller Module	ESP32-WROOM-32, 240 MHz, 4 MB Flash, WiFi 802.11 b/g/n
L298N Motor Controller Board	Dual H-bridge, 5 V onboard regulator, 2 A/ch
12 V DC Lift Motor	1.2 Nm, 80 RPM no-load, stall current 3.5 A
Lead Screw Assembly	8 mm diameter, 2 mm pitch, 150 mm travel
Homing Limit Switch	Snap-action lever, 5 A 250 VAC, IP40
Top Limit Switch	Snap-action lever, 5 A 250 VAC, IP40
Removable Needle Top Assembly	Dual tool clips; accepts Cytiva Allegro Style / Sanisure Fill4Sure

Station 2 — PlungerSet-80 (EA-PS-2024-00391)

Sub-assembly	Key Specification
ESP32 Controller Module	ESP32-WROOM-32, 240 MHz, 4 MB Flash, WiFi 802.11 b/g/n
L298N Motor Controller Board	Dual H-bridge, 5 V onboard regulator, 2 A/ch
12 V DC Lift Motor	1.2 Nm, 80 RPM no-load (interchangeable with Station 1)
Lead Screw Assembly	8 mm diameter, 2 mm pitch, 150 mm travel (interchangeable)
12 V Linear Insert Actuator	20 mm stroke, 50 N rated force, retract ~8 mm/s
SG90 Servo Feed Motor	5 V, 1.8 kg·cm stall torque, 2-stage 1:4 gear drive
Gear Drive Assembly	2-stage spur gear, 1:4 ratio; 10 mm pitch screw, 20 mm travel
Homing Limit Switch	Snap-action lever, 5 A 250 VAC, IP40
Plunger Magazine Assembly (std.)	Triangular-profile PLA channel, 4-plunger capacity

5 Consumables

The following consumable items are used in normal operation of the SPL-01 line. They are not tracked as fixed assets and are replenished by production personnel as part of normal batch preparation.

Item	Used at	Supplier	Notes
Sanisure Fill4Sure filling needle	Station 1	Sanisure	Single-use; lot number recorded in batch record
Cytiva Allegro Style filling needle	Station 1	Cytiva	Single-use; alternative to Fill4Sure
1 mL ISO rubber plunger — West 4432/50 Gray	Station 2	West Pharmaceutical Services	Loaded into magazine before each batch; up to 4 per load (standard)
1 mL ISO rubber plunger — Datwyler FM27	Station 2	Datwyler	Alternative validated plunger type
Replacement magazine channel (EA-PS080-SPARE-MAG)	Station 2	Elara Automation GmbH	Replace on visible cracking or deformation

6 Communication Overview

Station 1 and Station 2 use different communication protocols and connect to different supervisory endpoints. Both share the same physical WiFi access point and are on the same IP subnet. There is no direct station-to-station communication; coordination between the two stations is handled by the supervisory system.

Attribute	Station 1 — LinFill-120	Station 2 — PlungerSet-80
Protocol	MQTT 3.1.1	OPC UA TCP Binary
Port	1883 (prod: 8883 MQTTS)	4840
Device address	linfill120-EA-LF-2024-00472.local	plungerset80-EA-PS-2024-00391.local
Security mode	TLS 1.2 + username/password	SignAndEncrypt + X.509 certificate
State indication	Topic: {base}/filling/state	Node: ns=1;s=PlungerSet80.State
Control	Publish command string to .../cmd	Call method via OPC UA
Interface file	EA-MI-LF120-EN (PDF)	EA-NS-PS080.NodeSet2.xml

7 Environmental Conditions

Both stations share identical environmental ratings as they are built on the same platform. The shared ratings below apply to the complete SPL-01 line.

Ambient operating temperature	5 °C to 40 °C
Storage temperature	–20 °C to +60 °C
Relative humidity	20 – 80 % RH, non-condensing
IP rating	IP20 — indoor use only; no liquid ingress protection
Installation category	II (IEC 60664-1)
Pollution degree	2
Clean room compatibility	Not rated for ISO Class 7 or higher without additional enclosure

8 Document Register

The following documents make up the complete technical documentation package for the SPL-01 line. All documents are maintained by Elara Automation GmbH and are available from the manufacturer on request.

Document No.	Title	Format	Applies to
EA-LC-LF120-EN	LinFill-120 Lifecycle Information Document	PDF	SPL-01-001
EA-MI-LF120-EN	LinFill-120 MQTT Interface Specification	PDF	SPL-01-001
EA-LC-PS080-EN	PlungerSet-80 Lifecycle Information Document	PDF	SPL-01-002
EA-NS-PS080.NodeSet2.xml	PlungerSet-80 OPC UA NodeSet (complete address-space definition)	XML	SPL-01-002
EA-LN-SPL01-EN	SPL-01 Line Description & Equipment Register (this document)	PDF	SPL-01 Line

Document prepared by Elara Automation GmbH on behalf of the AAU SmartProduction Laboratory. For queries regarding this document contact support@elara-automation.de quoting document number EA-LN-SPL01-EN.